

# Information Loss and Recovery in SNN Training: Summary and Research Plan

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## Abstract

This document summarizes current observations about sources of information loss during spiking neural network (SNN) training, commonly used approximations and "fuzzy boundary" techniques that mitigate loss, quantitative metrics for measuring lost information, and proposed recovery mechanisms. It also proposes a comprehensive experimental research plan to systematically investigate information loss and recovery across major families of SNN training algorithms.

## 1 Survey Summary (LaTeX-format)

### 1.1 Algorithm families (observed in repository)

- BPTT (Backpropagation Through Time): full temporal gradient unfolding. Precise but memory-intensive ( $O(T)$ ) and subject to vanishing/exploding gradients.
- Three-factor / eligibility-trace methods (S-TLLR, TESS): local eligibility traces  $e_{ij}[t]$  combined with a learning signal  $\delta_i[t]$ ; online and local.
- Online/time-window approximations (OTTT, NDOT): methods cited in repository (bib entries present) for online temporal learning; full text needed to extract precise mechanisms.

### 1.2 Representative equations (from S-TLLR notes)

Membrane and spike model (discrete LIF):

$$u_i[t] = \gamma(u_i[t-1] - v_{th}y_i[t-1]) + \sum_j w_{ij}x_j[t],$$

$$y_i[t] = H(u_i[t] - v_{th}),$$

Three-factor update:

$$\Delta w_{ij}[t] = \delta_i[t] \cdot e_{ij}[t],$$

Eligibility trace (S-TLLR form):

$$e_{ij}[t] = \alpha_{\text{pre}} \sum_{\tau \leq t} \lambda^{t-\tau} x_j[\tau] \Psi(u_i[t]) + \alpha_{\text{post}} \sum_{\tau < t} \lambda^{t-\tau} \Psi(u_i[\tau]) x_j[t],$$

Recursive traces:

$$a_j^{\text{pre}}[t] = \lambda a_j^{\text{pre}}[t-1] + x_j[t], \quad b_i^{\text{post}}[t] = \lambda b_i^{\text{post}}[t-1] + \Psi(u_i[t]).$$

### 1.3 Qualitative categories of information loss

1. Temporal precision loss: due to time quantization/ binning, exponential decay ( $\lambda$ ) in eligibility traces, and temporal window truncation (truncated BPTT).
2. Spatial/representation precision loss: local learning rules and ANN-to-SNN conversions (rate/latency approximations) sacrifice precise spatiotemporal gradients.
3. Gradient direction and magnitude distortion: surrogate gradients ( $\Psi$ ), random feedback alignment, and truncated gradients perturb true gradient flow.
4. Discrete nonlinearity truncation: hard thresholding and hard reset discard continuous membrane potential information.

### 1.4 "Fuzzy boundary" / approximation techniques and their mechanisms

- Surrogate gradients ( $\Psi(u)$ ): smooth the non-differentiable spike function to recover usable gradient direction; amplitude and fine detail are biased but directionality preserved.
- Eligibility traces: accumulate decayed historical correlations to approximate long-range temporal credit assignment without full unfolding.
- Random/approximate feedback (Feedback Alignment): approximate backward signals with fixed or random matrices to preserve learning signal flow when exact backprop is infeasible.
- ANN→SNN conversion techniques: soften mapping (rate/latency smoothing, widened thresholds) to reduce discretization artifacts at cost of temporal detail.

### 1.5 Quantitative metrics for measuring information loss

- Mutual information  $I(X; S)$  between input signal  $X$  and spike train  $S$  (estimation requires careful high-dimensional estimators).

- Fisher information (or proxies): sensitivity of likelihood to parameter changes; truncated-gradient energy as an empirical proxy.
- Reconstruction / decoding error: MSE or SNR when reconstructing continuous inputs from spikes via an optimal decoder.
- Spike-timing distances: Victor–Purpura distance, van Rossum distance to quantify timing perturbations.
- Activity statistics: entropy of spike patterns, firing-rate coverage (proportion of time bins with spikes), empirical rank of activity matrix.
- Downstream task metrics (classification accuracy) as indirect indicators of preserved task-relevant information.

## 1.6 Recovery mechanisms (by stage)

### 1.6.1 During training

- Eligibility traces + three-factor rules (S-TLLR/TESS): keep decayed history to recover long-term credit.
- Dynamic/adaptive thresholds and soft-reset: reduce irreversible loss from hard resets.
- Surrogate gradients: restore gradient flow through discrete spikes.
- Latent-variable inference / variational methods: jointly infer continuous membrane states to recover hidden information.

### 1.6.2 After training

- Improved decoders (population decoding, nonlinear readouts) and temporal interpolation to reconstruct signals.
- Multi-pulse or wider-pulse encoding at readout to distribute information across channels and time.
- Generative denoising / diffusion or learned inverse models to reconstruct fine details lost by discretization.

### 1.6.3 Using constraints as reverse-information sources

- Interpreting truncated/approximate gradients as approximations to Fisher information and using them to adapt optimizer preconditioners.
- Variational frameworks treating spikes as observations and membrane potentials as latent variables, enabling posterior-based recovery.

## 2 Comprehensive Research Plan

### 2.1 Algorithm families to analyze

1. BPTT (full) and Truncated-BPTT baseline.
2. Surrogate-gradient methods (e.g., STBP, SLAYER – note: repository does not yet include these papers’ texts).
3. ANN-to-SNN conversion methods (rate/latency mapping).
4. Event-driven / online families (OTTT, NDOT) and three-factor eligibility rules (S-TLLR, TESS).

### 2.2 Experimental design (high level)

- Datasets: synthetic continuous trajectories (control over SNR and bandwidth) + standard SNN benchmarks (temporal MNIST variants, SHD, DVS datasets).
- Manipulations: vary temporal resolution (dt), input SNR, eligibility decay  $\lambda$ , surrogate gradient shape, truncated BPTT window length, and network redundancy.
- Measurements per condition: mutual information estimate  $I(X;S)$ , decoder MSE, spike-timing distances, Fisher-proxy, classification accuracy, and activity statistics.
- Protocol: for each method, keep training budget constant, train readout decoders (same architecture) on frozen spike outputs to measure recoverable information.

### 2.3 Distinguishing information loss vs compression

- Train an optimal decoder (unconstrained) on spike outputs. If decoder can recover input with low error, the process is compression; if recovery is poor despite decoder capacity, information is lost.
- Track mutual information and decoder generalization: compression often preserves task-relevant mutual information while reducing redundancy; irreversible loss decreases mutual information across all decoders.
- Use controlled perturbation experiments: inject small perturbations in input and measure sensitivity in spike statistics (if sensitivity preserved, information not lost).

## 2.4 Promising new directions

- Variational inference for membrane potentials as latent variables (VAE-style SNNs).
- Denoising/diffusion decoders to reconstruct high-frequency details from sparse spikes.
- Memory replay / eligibility consolidation mechanisms to re-inject long-term dependencies during online training.
- Architectures with reversible/near-reversible dynamics to reduce irreversible discretization loss.

## 3 Repository coverage and missing items

The repository contains detailed derivations for S-TLLR in [Notes/SNNInformationReconstruction.tex] and bib entries for NDOT, OTTT, and TESS under [Algorithm/]. However, full PDF/abstract text for NDOT, OTTT, TESS and other common methods (e.g., SLAYER, STBP, STDP-original papers) are not present in parsed plain text in the workspace. To map specific claims from those papers to the categories above, please provide either:

- PDF excerpts (plain text copied), or
- The paper PDFs and permission to perform a file-level read (note: current environment lacks a robust PDF-parsing tool).

## Next steps

1. If you provide paper abstracts/important paragraphs, I will annotate each paper with: which information-loss categories it triggers, whether it uses fuzzy/approximate mitigations, which metrics it reports, and whether it explicitly discusses "information loss".
2. Optionally I can expand this .tex into a full paper draft with figures and experimental code scaffold if you want me to run experiments (you must provide datasets and confirm compute constraints).